Jesús M. Rueda-Becerril, Ph.D.

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Professional Summary

Physicist (Ph.D. and M.Sc. in Physics) with ten years of research in computational astrophysics. Taught a graduate thermodynamics course and mentored three graduate researchers who co-authored papers with me. Teaching approach built on intuition, critical thinking, and collaborative problem-solving.

Teaching and Mentoring

Thermodynamics (graduate level) Jun 2018
Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo Substitute Instructor

Zachary Davis, graduate student 2018 – 2022
Department of Physics and Astronomy, Purdue University Mentoring

Amanda O. Harrison, graduate student 2018 – 2020
Department of Physics and Astronomy, Purdue University Mentoring

Hao Zhang, graduate student 2018 – 2019
Department of Physics and Astronomy, Purdue University Mentoring

Education

Universitat de València Jul 2017
Ph.D. in Physics, Excellent Cum Laude

Universidad Michoacana de San Nicolás de Hidalgo Sep 2011
M.Sc. in Physics

Universidad Autónoma del Estado de México Dec 2008
B.Sc. in Physics

Experience

Independent Contractor Jan 2026 – Present
Mercor

- Evaluate physics reasoning in AI model outputs for technical accuracy, identifying failure modes and providing structured feedback.

Postdoctoral Research Fellow Oct 2018 – Nov 2020
Purdue University

- Obtained and led a NASA-funded grant studying radiation from active galaxies.
- Mentored three graduate researchers; two are now postdoctoral researchers.
- Published 3 peer-reviewed papers.

Postdoctoral Research Associate Feb 2021 – Apr 2022
Rochester Institute of Technology

- Led an NSF-sponsored project upgrading a high-performance simulation code for modeling black hole binaries.
- Collaborated on a multi-institutional study of binary neutron star mergers, resulting in 2 publications.

Postdoctoral Research Fellow Jan – Sep 2018
Universidad Michoacana de San Nicolás de Hidalgo

- Developed a Python pipeline to process black hole simulation outputs from the GRTrans numerical code, generating SVM training data for radio image prediction.
- Built open-source data analysis and visualization tools in Python for radiative transfer phenomena in relativistic astrophysics scenarios.

Skills

Physics: Mechanics, electromagnetism, thermodynamics, relativity, statistical physics, astrophysics
Computing: Python, Fortran, C/C++, high-performance/parallel computing
Languages: English (fluent), Spanish (native), Nahuatl (basic)

Selected Publications (9 total, 3 first-author)

Full list available on request.